We thank the reviewers for their feedback. In the camera-ready we will fix the GIMDiffusion citation ([15]→[11]) and adopt "multi-chart geometry image" consistently. [R3 minor]

[R3] DMS-based end-to-end differentiability *to mesh* is the key contribution. Prior GI works (Omages, GIMDiffusion, GarmageNet) are differentiable only up to image-domain pixel losses; mesh extraction is a non-differentiable post-process, so 3D-domain supervision cannot reach the latent. DMS closes this gap, enabling Normal Rendering Loss to be computed in 3D and back-propagated to the VAE encoder/decoder (Supp. Sec. 6: "$\mathcal{L}_{\text{Normal}}$ can propagate gradients … *only because the DMS module is fully differentiable*"). Tab. 2 isolates the two effects. Representation alone (Occ→TSDF) gives CD $1.503 \rightarrow 0.595 \times 10^{-3}$ (-60%). Adding the Normal Loss—usable *only* via DMS—further improves NC $0.921 \rightarrow 0.961$ and JSD $2.169 \rightarrow 1.244 \times 10^{-3}$. TSDF + DMS + $\mathcal{L}_{\text{Normal}}$ is therefore a coupled contribution that enables 3D end-to-end learning, not a sum of standard components.

[R1] Isolating the effect of SD 1.5 VAE initialization. We use the SD 1.5 VAE only to accelerate convergence; trained on in-the-wild RGB, it carries no geometry-specific prior. We address R1's concern—"the gain may be confounded between representation and a stronger VAE"—in two steps. **(i)** Tab. 2 *fixes* the VAE init and varies only the representation: Occ→TSDF reduces CD $1.503 \rightarrow 0.595 \times 10^{-3}$ (-60%), demonstrating that the representation contributes independently. **(ii)** We are further running a from-scratch-vs.-SD-pretrained VAE comparison showing the two converge to comparable final performance (the latter only faster); this will be included in the supplementary.

[R1] Rationale for the unified DMS evaluation protocol. We agree that mesh extraction can affect outcomes. For this reason we apply the same extractor (DMS) to every baseline; on binary occupancy DMS produces a result equivalent to standard Marching Squares (Supp. Sec. 3), so no baseline is structurally disadvantaged. The protocol (Supp. Sec. 5) is designed to isolate *representation and compression* from mesh-extraction engineering—each baseline's native pipeline embeds different boundary heuristics that would otherwise mask the representation's own contribution. Under this protocol our method retains its advantage on both ABO and GarmageSet (Tab. 1). We are also preparing a mini comparison verifying the visual and numerical equivalence of DMS and standard MS on occupancy inputs, to be included in the supplementary.

[R3] Why no direct GIMDiffusion comparison. GIMDiffusion's code, weights, and preprocessed dataset are all unreleased, preventing same-condition reproduction. Nevertheless, its geometry pipeline (binary-mask GI + SD-initialized VAE + LDM; Collaborative Control is texture-only) is structurally equivalent to our "*Occ w/o $\mathcal{L}_{Normal}$*" ablation row. Against this baseline, our method (TSDF + $\mathcal{L}_{\text{Normal}}$) improves CD $1.503 \rightarrow 0.461 \times 10^{-3}$ ($3.3\times$) and JSD $4.539 \rightarrow 1.244 \times 10^{-3}$ ($3.6\times$, Tab. 2). We will make this relationship explicit in the camera-ready.

[R3] Speed vs. quality trade-off and structural limits of GarmageNet. GarmageNet's extreme 72-dim-per-panel compression yields fast diffusion but explicit quality cost: image-to-3D BCD 5.64 vs. ours 2.91×10^{-2} ($\sim 2\times$ worse, Tab. 5); reconstruction CD on GarmageSet 2.19 vs. 0.46×10^{-3} ($4.7\times$ worse, Tab. 1). Fig. 6 (main paper) further shows GarmageNet violating the input condition's structural attributes across multiple rows—garment-type switches (row 1: skirt→t-shirt; row 4: shorts→skirt), sleeve presence/length mismatches, and form breakdown with anomalous cuff in row 5. Its fixed per-panel structure also *precludes* application to variable-panel domains such as ABO (l. 274–275). Our method retains $\sim$1 s single-GPU latency while improving both quality and generality.

[R2] On UV-parameterization dependency. UV reliance is a constraint shared across the multi-chart GI family, ours included. In our target domains—garments and furniture—industry-standard UVs are typically available (ABO 7,900 commercial furniture; GarmageSet 14,801 professionally designed garments). As a side benefit, training on well-parameterized data yields meshes whose UVs are also clean, making them directly usable in simulation and texturing pipelines.

[R2] Resolution and latent compression. Supp. Sec. 1 shows reconstruction error converges beyond resolution 256—and even with $8\times$ spatial compression to a $32 \times 32 \times 4$ latent we outperform the uncompressed Omages ($64 \times 64 \times 4$) on CD/JSD/NC, because TSDF's redundant boundary encoding is robust to latent compression. Scaling to higher resolution simply enlarges the latent grid.

[R1] Seams in Fig. 6. Seams are a structural limitation of the multi-chart GI family, shared by all baselines. Our method substantially reduces boundary error in BCD (Tab. 5); complete removal is left as future work via dual-contouring integration (Sec. 5, Limitations).

[R1] TSDF is a 2D distance in UV-image space, not 3D (l. 139–141). For each pixel, the signed Euclidean distance to the nearest patch contour (positive inside a chart, negative outside), truncated at 15 pixels on the 1024×1024 grid (Sec. 3.1, Step 4). It is not a 3D point-to-surface distance. We will state this more clearly in the main text.

[R3] Non-cherry-picked ABO generation. ABO results are already in Tabs. 1, 4 and Fig. 5. To address cherry-picking concerns, Fig. 1 provides a latent-interpolation sequence between two unrelated samples on ABO—globally consistent, plausible furniture geometries appear throughout the entire path.

ECCV
#12200

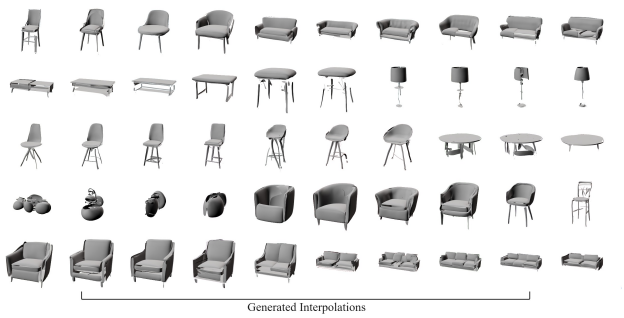

Figure 1. Non-cherry-picked latent interpolation on ABO. Linear interpolation in latent space between two unrelated samples produces globally consistent, plausible furniture geometries throughout the path.